Retail Supermarket Profit Margin Analysis

Retail Profitability & Margin Analysis

2024

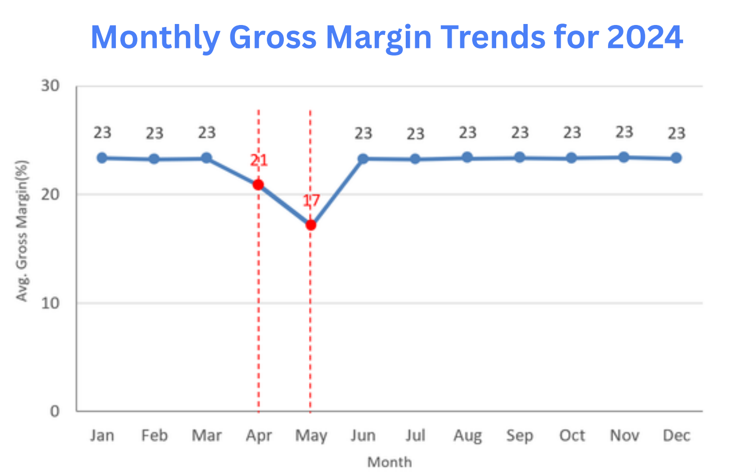
Business Context

A regional supermarket manages thousands of daily transactions across five major product categories. Sales patterns show significant seasonal fluctuations, particularly during the Eid al-Fitr period, when there is a surge in demand before the holiday and a sharp decline afterward. These conditions have the potential to cause inventory imbalances and operational inefficiencies, making analysis necessary to understand demand patterns and support more optimal decision-making.

Business Problem

The CFO stated that our gross profit margin dropped sharply in April–May 2024, even though the number of transactions remained relatively stable. The finance department is aware of this but has not yet been able to identify the cause. The data team has been asked to investigate and pinpoint the source of the problem. What caused this decline, why did it happen, and what steps can be taken to address the issue?

Monthly Gross Margin Trend

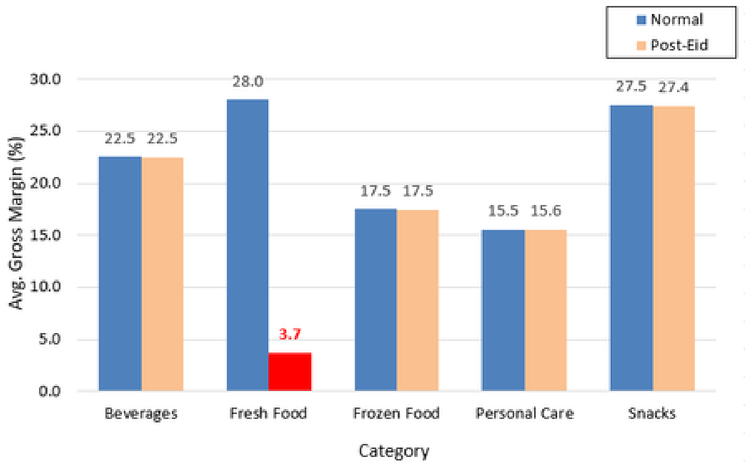


Insight

The gross margin fell sharply from 23% to 17% in April–May 2024, coinciding with the post-Eid al-Fitr period. The decline lasted only two months; the margin returned to normal in June and remained stable for the rest of the year. This suggests that the cause was seasonal, rather than a structural decline in business performance.

Margin Comparison by Category: Normal vs. Post-Eid

Comparison of Gross Margin by Category: Normal vs. Post-Eid



Insight

Fresh Food was the only category affected, with margins dropping from 28% to 3.7% after Eid. Other categories remained stable, confirming that the decline was category-specific rather than a broad-based decline across the business.

Order Quantity vs. Quantity Sold per Season (Fresh Food)

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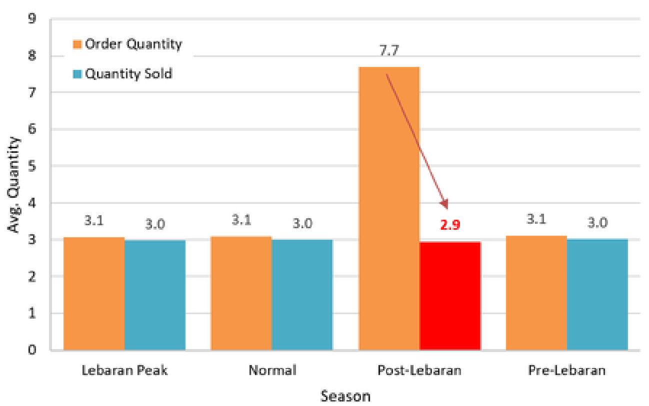

Insight

After Eid, order volume surged dramatically to 7.7, but sales volume was only 2.9, meaning that the gap between order quantity and quantity sold in the post-Eid period was quite significant, resulting in excess inventory, which was likely the cause of the sharp decline in Fresh Food margins during that period. The fact that orders were 2.5 times higher than usual was the main factor behind the increase in shrinkage rates and reactive discounts during that period.

The Factors Contributing to the Decline in the Gross Profit Margin

Factors Behind the Fresh Food Margin Squeeze During Post-Eid Period

Factor	Normal	Post-Eid	Impact
Shrinkage Rate	3.9%	11.1%	The rate of waste has 3x compared to normal levels
Avg. Discount	0%	19.8%	Big discounts to clear out inventory, not a planned strategy.
Order Qty **vs** Sold	Proporsional	Over-order 2.5x	Inventory is stacked up after Eid al-Fitr
Supplier Cost Index	1	1.080	The purchase price increased by 8%

Analytical Conclusion

The gross profit margin declined not because of sluggish sales, but because the Fresh Food division faced several issues during the post-Eid period: **(1)** shrinkage surged due to over-ordering, **(2)** reactive discounts had to be offered to clear out fresh food inventory nearing its expiration date, and **(3)** purchase prices from suppliers rose. The root cause was that the purchasing team failed to adjust order quantities to account for the post-Eid decline in demand. This combination of three factors contributed to a 6% decline in margins for Fresh Food during the post-Eid period.

Recommendation

1. Implement demand-based ordering for fresh food after Eid. Use historical post-Eid sales data as the baseline order quantity, with buffer stock adjusted based on the shelf life of fresh food products.
2. Set a maximum limit on reactive discounts for fresh food. A policy must be established stipulating that promotional discounts on fresh food must not exceed 5% unless special circumstances require otherwise.
3. Renegotiate supplier contracts. Supplier price increases of 8% post-Eid. The procurement team should negotiate flat-rate or volume-flexible contracts specifically for peak demand periods.

Information:

Dataset: Retail Supermarket